Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Extremas of functions continued

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Extremas of functions continued
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Extremas of functions continued

1 Extremas of functions continued

Study focus: Extremas of functions continued. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000206 · EV0000207

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Find the highest temperature on the surface of the unit sphere.

Key details

- because it's already in the xy plane.
- But do we know how to find the extremas for a function of three independent variables?
- That's a problem I have.
- I can write z square as 1 minus y square minus x square.
- x y into for z square I can substitute 1 minus x square minus y square.
- You differentiate that with respect to x partially and equate it to 0.
- I will get 400 y is a constant, I do not need to worry, derivative of x is 1, derivative of x cube is 3 x square and derivative of x y square is y square.
- If I equate this to 0,.
- Then I get 1 minus 3x square minus y square to be 0.
- That's a derivative.
- And if you equate this to zero you will get this equation.
- They are like simultaneous equations.
- you subtract them together.
- If you subtract I'll get minus two x square and then plus two y square equal to zero.
- If x is plus or minus y,.
- What w what is the value of x and y that I get if I substitute as x is equal to plus or minus y there.
- Let me substitute in either this equation or this equation, it doesn't matter.
- Or I get 1 is equal to 4 x square or x square equal to 1 by 4.
- But also what we got is y equal to 0 or this.

- Actually this should be x equal to 0, not y equal to 0.
- x is equal to minus half and y can be minus half.
- What we need to do is we just need to c get a combination of these things.
- You get these many points.
- Either it could be 0 half comma half, half comma minus half, minus half comma half and minus half comma minus half.
- That means these are the points at which there is a possibility for the function to have an extrema.
- I don't know whether it will be a maximum or a minimum or a saddle point, I don't know.
- So next thing what we need to do is we need to find the second order partial derivatives.
- And for half comma half it is positive, for minus half comma half also it is positive.
- And so you can calculate what is the maximum value that it can attain and minimum value it can attain by substituting these values, right.

Example or application from the lecture

- First thing is so that means they have given a sphere which might look something like that.
- And they have given the temperature function t of x y z as 400 x y z square which varies with the par with the spatial coordinates.
- So what I will do is so this is a constraint that is given to me, that means you need to find the temperature in space, find the highest temperature on the surface of the sphere.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000206 · EV0000207 · EV0000208 · +1 source units in pack.json

Core explanation · 05:00–09:59

maximum at two points.

Key details

- Can we conclude just by taking this?
- Just by taking this condition?
- I need to check along with that R is less than zero.
- I can erase the things that are there here.
- The function is this.
- We got F x already as four hundred y into one minus y square minus three x square.
- What is f x?
- I need to differentiate this which will give me minus 6x.
- Because it satisfies this condition, we say that the T is maximum at these two points on the surface of a sphere.
- And if you want to calculate what is that maximum value, substituting four hundred so we have four hundred into x.
- into y into z square.
- But z square is actually 1 minus x square minus y square.
- That means the maximum temperature that is attained on the surface of a sphere is fifty.
- So this is one method wherein we actually eliminate and then we substitute.

- This is more of a practical problem, but I felt like you will appreciate this because you will understand how to solve a problem.
- It says a container with an open top, let me underline whatever you need to be careful about is to have a ten meter cube capacity.
- and be made of thin sheet of metal.
- Calculate the dimensions of the box if it is to use the minimum possible amount of metal.
- So it's an open top means what?
- Because I am worried about the surface area.
- I hope it's clear why didn't I prefix this with two is because my top is open here as you can see this is not closed so because it's an open top box.
- That's a wall.
- I hope you understood how I wrote this.
- This is because I need to minimize the possible amount of metal that is required for me to made it is made out of thin.
- I cannot handle three independent variables, right?
- Like we did in the previous case, we use the equation of the sphere.
- We are going to eliminate z.
- I am doing little simplification here.
- So this is the function which I need to minimize because I want the minimum possible amount of metal to be used.
- Now what are the condition for this to be minimum?
- area is equal to 2xy plus 20 divided by x plus twenty divided by y and that's what we got.
- What are the conditions for us to minimize this?
- So the first thing is I need at what points it can be minimum.
- Ax will be 2 y minus 20 by x square, right?
- Equate this to 0.
- What do I get if I equate this to zero?

Example or application from the lecture

- So at any given point because of this negative, okay, if both x and y are positive, or both x and y are negative, I'll get f of x to be negative and that's what I want, right?
- Let's do another problem.
- And of course it's a rectangular box, so they have given XYZ to be ten.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000210 · EV0000211 · EV0000212 · +1 source units in pack.json

Core explanation · 10:00–14:59

I can take out y comma two comma and I can divide two out through out by two.

Key details

- Is that fine?
- So this is the first equity that I get y is equal to twenty by or ten by x square.
- What's a y?

- I will get 2x again minus 20 by y square equal to 0.
- If I equate this to 0, I will get x is equal to 10 by y square.
- Where should I put that to solve, right?
- I need to solve for x and then y and then z to get the critical points.
- The only equation that I know is XYZ equal to ten.
- I got x y is equal to ten by x square and x is equal to ten by y square.
- What is the next thing I know?
- I know that I have to find a x a y minus a x y the whole square must be greater than zero.
- Alright, so I have these two equations now y is equal to 10 by x square and x is equal to 10 by y square.
- Can I substitute one in another?
- I have y is equal to 10 by x square.
- Or I can eliminate this.
- Can we solve this?
- I'll get y minus y to the power of four by ten to be equal to zero.
- Is it fine?
- So that means I can take out y common and then I get 1 minus y cube by 10 equal to 0.
- What do we get when we solve this equation?
- I'll get y equal to 0.
- as the first value or so the other equation that I have is 1 minus 1 minus y cube by 10 equal to 0.
- Y is equal to zero doesn't make sense for me.
- And so how do I find x now?
- I know that x is equal to ten by y squared.
- And of course the y equal to 0 doesn't work because if y becomes 0 x tends to infinity.
- That's another way to eliminate that.
- And how do I get z?
- Z is equal to 10 by x y.
- This is 10 divided by 10 to the power of 2 by 3 or this is 10 to the power of 1 minus 2 by 3 which is 10 to the power of 1 by 3 again.
- Is it fine?
- And the we just substituted that.
- We got x is equal to y is equal to z.
- But how do we check that this will m this is the minimum value or the minimum amount of metal that I can get, right?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000214 · EV0000215 · EV0000216 · +1 source units in pack.json

Core explanation · 15:00–19:59

20 by x square and derivative of 1 by x square will be 2 by x cube with a minus.

Key details

- And a y will be same thing, which is 40 by y cube.

- And then I need another axy.
- What are the condition we have?
- Ax x into ay minus a x y the whole square.
- And what happens to this?
- A x we got the value right.
- What is the value that we got?
- 40 by x cube.
- So that means this is greater than 0.
- And what is Ax?
- You can observe that for all the points that we got for x, y and z, a x is always greater than 0.
- That means the dimensions of the box so that the minimum of metal sheet is used will be x is equal to cube root of ten.
- It's a basically a cube and y is equal to cube root of ten and z is equal to cube root of ten.
- I hope the problem is clear.
- in the next session we will see another method wherein we don't have to eliminate.
- See the problem with us we had like we have to eliminate one of the variables to make it into a function of two independent variables.
- It might not be as simple as this in every case, right.
- So we'll see another method which will help you to minimize that so that how to solve the problem.

Example or application from the lecture

- So, the condition that we got we got the critical points now let's check the condition.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000218 · EV0000219 · EV0000220 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Extremas of functions continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00054	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00055	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...
SRC00056	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00057	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare...

C Revision and retrieval practice